

L-functions and periods of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$
 Comments: (1) Working draft, currently restricted to $\dim S_k \leq 1$,
 (2) the properly meromorphic MF case at very preliminary stage

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Abstract

For any $f \in S_{12}^!$ on $\mathrm{SL}_2(\mathbb{Z})$ (the weakly holomorphic forms of weight 12 with vanishing constant Fourier coefficient; this includes the unique normalised holomorphic cusp form $\Delta \in S_{12} \subset S_{12}^!$) we exhibit an explicit linear functional $f \mapsto (\omega^+(f), \omega^-(f)) \in \mathbb{C}^2$ given by integration of f along two finite line segments in the upper half-plane against universal $\mathbb{Q}$ -polynomial kernels. Specialising to Δ recovers Eichler–Shimura’s classical periods $\omega^\pm(\Delta)$; specialising to the unique $\hat{\Delta} \in S_{12}^!$ with $\hat{\Delta}(\tau) = q^{-1} + O(q^2)$ produces Brown’s quasi-periods $\eta^\pm(\hat{\Delta})$. The construction is the explicit Bernoulli-operator unpacking of the Diamantis–Rolen finite-endpoint cocycle; the functional is basepoint-independent. Replacing the $\mathbb{Q}$ -polynomial T -kernel by a Hurwitz-zeta function of τ promotes the integer- s formula to a finite-contour expression for the completed L -function $\Lambda^*(f, s)$ at all complex s , which agrees established L -function at all values in the critical strip. The finite-contour formulation extends without modification to meromorphic modular forms with interior poles, with a wrinkle: τ_0 -invariance is broken by discrete wall-crossing jumps that encode the residues of f at its interior poles, suggesting the natural starting point for an L -function theory of generic meromorphic modular forms.

1 Introduction and main result

The period polynomial of a weight- k holomorphic cusp form $f \in S_k$ on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is the polynomial

$$r_f(X) := \int_0^{i\infty} f(\tau) (\tau - X)^{k-2} d\tau \in V_{k-2} \otimes \mathbb{C},$$

which decomposes into even and odd parts $r_f^\pm$ whose coefficients are $\mathbb{Q}$ -rationals multiples of $\omega^\pm(f) (2\pi i)^\bullet$, with the periods $\omega^\pm(f)$ in turn expressible as $\mathbb{Q}$ -multiples of critical L -values $L(f, s)$. The Eichler–Shimura picture realises $\omega^\pm(f)$ as $\mathbb{C}$ -linear functionals of f via a contour integral, and the resulting coordinates carry deep arithmetic.

For weakly holomorphic $f \in S_k^!$ with a cusp pole, the integral $\int_0^{i\infty}$ diverges and the classical construction fails. Nevertheless Brown [1] defines and numerically computes *quasi-periods* $\eta^\pm(f)$ for the unique form $\hat{\Delta} \in S_{12}^!$ satisfying $\hat{\Delta}(\tau) = q^{-1} + O(q^2)$.

To implement Brown’s construction, we use the Diamantis–Rolen finite-endpoint cocycle [2]: replacing the divergent contour $0 \rightarrow i\infty$ with a finite segment $\gamma^{-1}\tau_0 \rightarrow \tau_0$ in $\mathbb{H}$ produces a genuine 1-cocycle C^f of Γ whose cohomology class encodes $\omega^\pm(f)$ for holomorphic f and $\eta^\pm(f)$ for weakly holomorphic f , on the same footing. Brown’s $\omega^\pm$ and $\eta^\pm$ thus come from a single procedure. The present note unpacks this procedure to the level of an explicit linear functional, fully transparent at the coefficient level.

Conventions. Fix $k = 12$, $n = k - 2 = 10$. Let $V_n = \bigoplus_{i+j=n} \mathbb{Q}X^iY^j$ with the right Γ -action $(X, Y)|_\gamma = (aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Write $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. We write $[P]_{X^{10-j}Y^j}$ for the coefficient of $X^{10-j}Y^j$ in $P \in V_{10}$, and set $q = e^{2\pi i\tau}$. We use the convention that $S_k^!$ denotes the weakly holomorphic forms of weight k with vanishing constant Fourier coefficient, so $S_k \subset S_k^!$.

Zagier's space $W_{12} \subset V_{10}$ of admissible period polynomials at weight 12 is 3-dimensional, with the standard basis

$$\begin{aligned} P_0(X, Y) &= X^{10} - Y^{10}, \\ P_1(X, Y) &= X^8Y^2 - 3X^6Y^4 + 3X^4Y^6 - X^2Y^8, \\ P_2(X, Y) &= 4X^9Y - 25X^7Y^3 + 42X^5Y^5 - 25X^3Y^7 + 4XY^9, \end{aligned} \tag{1}$$

splitting as $W_{12} = W_{12}^+ \oplus W_{12}^-$ with $W_{12}^+ = \langle P_0, P_1 \rangle$ (the 2-dimensional even-in- X sector, containing the Eisenstein direction P_0) and $W_{12}^- = \langle P_2 \rangle$ (the 1-dimensional odd-in- X cuspidal sector) [5]. Brown's basis at S [1, §8.2] is $P_S^+ = -\frac{36}{691}P_0 + P_1$ and $P_S^- = P_2$.

The two main results. First, for each interior monomial $j \in \{1, 2, \dots, 9\}$ let ε_j denote the parity sign — if j is odd, + if j is even — and let $c_S^j := [P_S^{\varepsilon_j}]_{X^{10-j}Y^j} \in \mathbb{Q}$ denote the coefficient of $X^{10-j}Y^j$ in Brown's basis vector of the appropriate parity.

Second, we extend this into an L-function to hold for a generic meromorphic modular, by generalizing off of $s \in \{1, \dots, k-1\}$ to a function in the entire s -plane in a natural way. We prove that these results exactly reproduce those in [3]. We note that the construction also includes the case of L-functions associated with meromorphic modular forms that are regular at cusps, introduced in [4]. Further, we emphasize that these newer L-function derivations naturally follow from first principles and do not depend on any regularization at all.

Theorem 1.1 (Periods and quasi-periods as a linear functional). *Focus on k such that $\dim(S_k) \leq 1$. Here, there exist universal polynomials $k_S^j, k_T^j \in \mathbb{Q}[\tau]$ of degree $\leq k-2$ in τ , indexed by $j \in \{1, \dots, k-2\}$ and, for the case of $k = 12$ are given explicitly in Table 1 below, with the following property. For any $\tau_0 \in \mathbb{H}$ and any $f \in S_k^!$,*

$$\omega^{\varepsilon_j}(f) = \frac{(2\pi i)^{k-1}}{c_S^j(-1)^j} \binom{10}{j} \left[\int_{-1/\tau_0}^{\tau_0} f(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} f(\tau) k_T^j(\tau) d\tau \right]. \tag{2}$$

The value $\omega^{\varepsilon_j}(f)$ is independent of the basepoint τ_0 and of the choice of interior monomial j (within its parity sector). Both contour integrals are finite for every $f \in S_k^!$, with no regularisation, analytic continuation, or limit required.

Theorem 1.2 (Complex- s finite-contour L-function for Δ). *For all $s \in \mathbb{C} \setminus \{1, k-1\}$ and all $\tau_0 \in \mathbb{H}$, and for every $f \in S_k^!$ where $\dim(S_k) \leq 1$, define*

$$L_f^*(s) = i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \tag{3}$$

where the Hurwitz-zeta T -kernel is

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1). \tag{4}$$

This $L_f^(s)$ is the L-function for the classical case of $f \in S_k$ and equals the L-functions previously defined in [3] and [4]. However, it does so without any need to regularize divergences. Further, $L_f^*(s)$ satisfies the functional equation $L_f^*(s) = i^k L_f^*(k-s)$.*

Theorem 1.1 is the analog of the classical $\omega^\pm \propto L(f, s) (2\pi i)^\bullet$ formula in finite-contour DR coordinates, valid uniformly for holomorphic and weakly holomorphic f . We prove it in §§2–6: existence of the kernels via the Bernoulli operator (§4), basepoint-independence via an explicit coboundary identity (§6), j -independence via the lockstep structure of W_{12} . Two specialisations of immediate interest:

Corollary 1.3 (Classical periods $\omega^\pm(\Delta)$). *For $f = \Delta \in S_{12} \subset S_{12}^!$, the unique normalised holomorphic cusp form, (2) reproduces the Eichler–Shimura periods. At $\tau_0 = 0.3 + 1.2i$ all nine extractions agree (relative residual $\sim 10^{-24}$) on*

$$\omega^+(\Delta) = -68\,916\,772.80959519475431\dots, \quad \omega^-(\Delta) = -5\,585\,015.379310401866877\dots i,$$

matching Brown [1, §8.3] to all 20 reported digits.

Corollary 1.4 (Quasi-periods $\eta^\pm(\hat{\Delta})$). *For $f = \hat{\Delta} = \Delta \cdot (J^2 + 24J - 393444) \in S_{12}^!$, (2) reproduces Brown’s quasi-periods. At $\tau_0 = 0.3 + 1.2i$, all nine extractions agree (relative residual $\sim 10^{-17}$) on*

$$\eta^+(\hat{\Delta}) = 127\,202\,100\,647.1770947773\dots, \quad \eta^-(\hat{\Delta}) = 10\,276\,732\,343.6491327508\dots i,$$

matching Brown [1, §8.3] to all 20 reported digits.

The integrand and contour structure are *identical* in Corollaries 1.3 and 1.4; only the modular form f under the integral sign changes. This is the precise sense in which the DR construction places $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ on the same footing.

Corollary 1.5 (Uniform applicability across $\mathbb{C}[J] \cdot \Delta$). *For any $R \in \mathbb{C}[J]$ with $a_{R\Delta}(0) = 0$, formula (2) computes $\omega^\pm(R \cdot \Delta)$ via the same universal kernels. In particular, the entire $\mathbb{C}$ -span of products $R(J) \Delta$ (with vanishing constant term) maps linearly into $H_{\text{cusp}}^1(\Gamma; V_{10} \otimes \mathbb{C}) = \mathbb{C}[P^+] \oplus \mathbb{C}[P^-]$, landing on the pair $(\omega^+, \omega^-) \in \mathbb{C}^2$ via (2).*

Why this matters. A “period” is, in classical usage, the integral of an algebraic differential along a cycle. Eichler–Shimura realises $\omega^\pm(\Delta)$ this way: as Mellin transforms of Δ on $[0, i\infty]$, packaged into a polynomial in V_{10} . Theorem 1.1 preserves the period-as-integral picture but on a finite contour in $\mathbb{H}$, with universal polynomial kernel weights k_S^j, k_T^j in place of the bare τ^j moments. For holomorphic Δ , taking $\tau_0 \rightarrow i\infty$ collapses the T -segment, sends the S -segment to $[0, i\infty]$, and reduces the kernels to bare monomials — recovering classical Mellin (Lemma 5.3). For weakly holomorphic $\hat{\Delta}$, the limit is unavailable, but at every finite $\tau_0 \in \mathbb{H}$ the formula and its value are well-defined. “Quasi-periods” are honestly periods in the same geometric sense as the classical ones: integrals of a modular form along explicit cycles in $\mathbb{H}$, against explicit polynomial kernels, with the contour bounded away from the cusp. The basepoint τ_0 — not analytic continuation, not a non-holomorphic regulator — is what tames the cusp pole.

Outline. §2 introduces the Diamantis–Rolen cocycle and its moment expansion; §3 introduces the coboundary modification by a polynomial P satisfying $P|_T - P = C_T^f$ (existence, uniqueness, Bernoulli closed form); §4 establishes the universal kernels of Theorem 1.1 and presents Table 1; §5 verifies the formula numerically for Δ and $\hat{\Delta}$ and shows it produces an exponentially convergent q -series for $\omega^\pm$; §6 interprets $\omega^\pm$ as H_{cusp}^1 coordinates and proves the basepoint and j -independence; §7 contrasts with BFK’s regularisation; §9 sketches the extension to fully meromorphic forms with interior poles and offers a gauge-theoretic analogy as a thread for future work.

2 The Diamantis–Rolen cocycle and its moment expansion

Definition 2.1 (Diamantis–Rolen finite-endpoint cocycle). For $f \in S_{12}^!$ and $\tau_0 \in \mathbb{H}$, define $C^f: \Gamma \rightarrow V_{10} \otimes \mathbb{C}$ by

$$C_\gamma^f(X, Y) := (2\pi i)^{11} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^{10} d\tau, \quad (5)$$

the contour being the straight line segment from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$.

Lemma 2.2. C^f is a 1-cocycle of Γ : $C_{\gamma_1\gamma_2}^f = C_{\gamma_1}^f|_{\gamma_2} + C_{\gamma_2}^f$ for all $\gamma_1, \gamma_2 \in \Gamma$.

Proof. Path-additivity: $\int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\tau_0} = \int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\gamma_2^{-1}\tau_0} + \int_{\gamma_2^{-1}\tau_0}^{\tau_0}$. The substitution $\tau = \gamma_2\sigma$ in the first integral, combined with weight-12 modularity of $f(\tau) d\tau$ (transforming with weight 10 under $\sigma \mapsto \gamma_2\sigma$, exactly cancelling the weight-10 slash on $(X - \tau Y)^{10}$), identifies it with $C_{\gamma_1}^f|_{\gamma_2}(X, Y)/(2\pi i)^{11}$. $\square$

A 1-cocycle of Γ is determined by its values at the generators S, T . Define the *moment integrals*

$$N_j(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau, \quad M_j(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^j d\tau, \quad j = 0, 1, \dots, 10. \quad (6)$$

Expanding $(X - \tau Y)^{10}$ binomially in (5) gives

$$C_S^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} N_j X^{10-j} Y^j, \quad (7)$$

$$C_T^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} M_j X^{10-j} Y^j. \quad (8)$$

For $f \in S_{12}^!$ the q -coefficients $a_f(n)$ grow at most as $e^{4\pi\sqrt{n}}$ [1], while $|q|^n = e^{-2\pi n \operatorname{Im}(\tau)}$ decays super-exponentially in n for any $\tau \in \mathbb{H}$. The Fourier series therefore converges absolutely on every compact subset of $\mathbb{H}$, and the moment integrals (6) are unambiguously finite for any $\tau_0 \in \mathbb{H}$.

Remark 2.3. The cuspidality input $a_f(0) = 0$ (built into the definition of $S_{12}^!$) enters in exactly one place: it makes $M_0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f d\tau = a_f(0)$ (by periodicity of f), so vanishing of $a_f(0)$ is equivalent to vanishing of the X^{10} coefficient of C_T^f . This is the solvability condition for the coboundary equation in §3.

3 Cuspidality and the coboundary polynomial

The cocycle C^f encodes more data than $\omega^\pm(f)$. To extract the periods we replace C^f by a cohomologous *cuspidal* cocycle whose value at T vanishes. The price is a coboundary modification by $P \in V_{10} \otimes \mathbb{C}$ satisfying

$$P|_T - P = C_T^f. \quad (\star)$$

The modified cocycle

$$C_\gamma'^f := C_\gamma^f - (P|_\gamma - P) \quad (9)$$

is cohomologous to C^f and satisfies $C_T'^f = 0$.

Proposition 3.1 (Existence and uniqueness of P). *The operator $\delta_T: V_{10} \rightarrow V_{10}$, $P \mapsto P|_T - P$, has $\ker(\delta_T) = \mathbb{C} \cdot Y^{10}$ and $\text{im}(\delta_T) = \{Q \in V_{10} : [Q]_{X^{10}} = 0\}$. Hence $(\star)$ is solvable iff $[C_T^f]_{X^{10}} = 0$, which by Remark 2.3 is equivalent to $a_f(0) = 0$ (automatic for $f \in S_{12}^!$). The solution P is unique modulo $\mathbb{C} \cdot Y^{10}$.*

Proof. T acts on V_{10} as $X \mapsto X + Y$, $Y \mapsto Y$, fixing only powers of Y , hence $\ker(\delta_T) = \mathbb{C} \cdot Y^{10}$. Writing $P = \sum_{j=0}^{10} p_j X^{10-j} Y^j$ and expanding $(X + Y)^{10-j}$ binomially, the $X^{10-m} Y^m$ coefficient of $P|_T - P$ is $\sum_{j=0}^{m-1} \binom{10-j}{m-j} p_j$. This is upper-triangular in m : the $m = 0$ row is identically 0 (requiring $[C_T^f]_{X^{10}} = 0$), and rows $m = 1, \dots, 10$ determine $p_0, \dots, p_9$ in turn, with p_{10} unconstrained. $\square$

We fix the convention $p_{10} = 0$; alternative choices shift $C_S'^f$ by $c(X^{10} - Y^{10})$ (since $Y^{10}|_S - Y^{10} = X^{10} - Y^{10}$), a coboundary in V_{10} leaving the cohomology class and the periods invariant.

Proposition 3.2 (Bernoulli closed form for P). *With $T = e^{Y\partial_X}$ acting on V_{10} as a translation operator, the unique solution of $(\star)$ with $p_{10} = 0$ is*

$$P(X, Y) = \frac{1}{Y} \int_0^X C_T^f(t, Y) dt + \sum_{k=1}^{10} \frac{B_k}{k!} Y^{k-1} \frac{\partial^{k-1} C_T^f}{\partial X^{k-1}}(X, Y), \quad (10)$$

where B_k are Bernoulli numbers ($B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $\dots$; $B_{2\ell+1} = 0$ for $\ell \geq 1$). The series terminates at $k = 10$, with at most six nonzero Bernoulli terms.

Proof. Formally $\delta_T = e^{Y\partial_X} - 1$ as an operator, so $\delta_T^{-1} = \sum_{k \geq 0} B_k (Y\partial_X)^{k-1} / k!$ by the Bernoulli generating identity $z/(e^z - 1) = \sum_k B_k z^k / k!$. The $k = 0$ summand is $(Y\partial_X)^{-1} = \partial_X^{-1} / Y$ (the antiderivative in X divided by Y), giving the integral term; the $k \geq 1$ summands give the differential terms. On a polynomial of finite X -degree the operator series terminates. Agreement of (10) with the upper-triangular recursion in the proof of Proposition 3.1 is verified to relative residual $\sim 10^{-42}$ at 45-digit working precision in [10]. $\square$

Substituting (8) into (10) expresses each p_l as an explicit $\mathbb{Q}$ -linear combination of the moments $M_1, \dots, M_{10}$:

$$p_0 = -(2\pi i)^{11} M_1, \quad p_1 = 5(2\pi i)^{11} (M_1 + M_2), \quad p_2 = -(2\pi i)^{11} \left(\frac{15}{2} M_1 + \frac{45}{2} M_2 + 15 M_3 \right), \quad (11)$$

with $p_3, \dots, p_9$ following the same pattern.

Definition 3.3 (Modified cocycle value $C_S'^f$). *With P as in Proposition 3.2 and using $P|_S(X, Y) = P(Y, -X)$:*

$$C_S'^f(X, Y) := C_S^f(X, Y) - [P(Y, -X) - P(X, Y)].$$

This is a polynomial in $V_{10} \otimes \mathbb{C}$ whose coefficients are explicit linear combinations of the moments $\{M_j, N_j\}_{j=0}^{10}$ of f , with rational/Bernoulli weights and overall factor $(2\pi i)^{11}$.

By construction $C_T'^f = 0$, so $C_S'^f$ is the S -value of a cuspidal 1-cocycle and admits Brown's decomposition

$$C_S'^f = \omega^+(f) P_S^+ + \omega^-(f) P_S^- + c(f) (X^{10} - Y^{10}), \quad (12)$$

with the $c(f)$ ambiguity absorbing the $p_{10} = 0$ convention.

4 The universal polynomial kernels

Definitions 2.1, 3.3 and Proposition 3.2 express $C_S'^f$ as an explicit $\mathbb{Q}$ -linear combination of the moments $\{N_j, M_j\}_{j=0}^{10}$ of f . Reading off interior-monomial coefficients $[C_S'^f]_{X^{10-j}Y^j}$ from (12) and dividing by c_S^j gives $\omega^{\varepsilon_j}(f)$ as a $\mathbb{Q}$ -linear functional of the moments. Each N_j (resp. M_j) is an integral of f against τ^j on the S -segment (resp. T -segment), so chaining gives an integral of f against an explicit polynomial in τ .

Theorem 4.1 (Universal kernel polynomials). *For each interior $j \in \{1, 2, \dots, 9\}$, define*

$$k_S^j(\tau) := (-1)^j \binom{10}{j} \tau^j, \quad (13)$$

$$k_T^j(\tau) := \sum_{r=1}^{10} [\beta_{j,r} - (-1)^j \beta_{10-j,r}] \tau^r, \quad (14)$$

where $\beta_{l,r} \in \mathbb{Q}$ are the Bernoulli-operator weights from (10) computing the $X^{10-l}Y^l$ coefficient of P from the moment M_r ,

$$\beta_{l,r} = \frac{(-1)^r \binom{10}{r} B_{l-r+1} (10-r)!}{(l-r+1)! (10-l)!} \quad \text{for } 0 \leq l \leq 9, 1 \leq r \leq l+1, \quad (15)$$

and $\beta_{l,r} = 0$ otherwise (in particular $\beta_{10,r} = 0$ for all r , encoding $p_{10} = 0$). With these kernels, identity (2) of Theorem 1.1 holds for each interior j .

Proof. By Definition 3.3, $[C_S'^f]_{X^{10-j}Y^j} = [C_S^f]_{X^{10-j}Y^j} - [P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j}$. The first term is $(2\pi i)^{11} (-1)^j \binom{10}{j} N_j$ by (7); since $N_j = \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau$, the S -segment contribution to $\omega^{\varepsilon_j}(f) = [C_S'^f]_{X^{10-j}Y^j} / c_S^j$ is $(2\pi i)^{11} / c_S^j \cdot \int_{-1/\tau_0}^{\tau_0} f(\tau) k_S^j(\tau) d\tau$ with k_S^j as in (13).

For the second term, write $P = \sum_l p_l X^{10-l} Y^l$ and $P(Y, -X) = \sum_l p_l (-1)^l X^l Y^{10-l}$. The $X^{10-j}Y^j$ coefficient of $P(Y, -X)$ is $(-1)^{10-j} p_{10-j} = (-1)^j p_{10-j}$ (since $(-1)^{10} = 1$), so $[P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j} = (-1)^j p_{10-j} - p_j$. Each p_l is a $\mathbb{Q}$ -linear combination of moments $p_l = (2\pi i)^{11} \sum_r \beta_{l,r} M_r$ via (10); direct expansion gives the formula (15). Therefore

$$[P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j} = (2\pi i)^{11} \sum_{r=1}^{10} [(-1)^j \beta_{10-j,r} - \beta_{j,r}] M_r,$$

and the T -segment contribution to $\omega^{\varepsilon_j}(f)$ is $-(2\pi i)^{11} / c_S^j \cdot \int_{\tau_0}^{-1/\tau_0} f(\tau) Q(\tau) d\tau$ where $Q(\tau) = \sum_r [(-1)^j \beta_{10-j,r} - \beta_{j,r}] \tau^r$. The negative sign flips this into $k_T^j(\tau)$ as in (14). Adding the S - and T -contributions gives (2). Independence of τ_0 and j is Theorem 6.1 below. $\square$

The kernels are universal: they depend only on j , not on f or τ_0 . Computing them via sympy and the Bernoulli formula (10)–(15) [12] gives Table 1.

Remark 4.2. *The S -kernel $k_S^j(\tau)$ is a pure monomial of degree j — it comes directly from the bare τ^j moment N_j . All complexity sits in the T -kernel $k_T^j(\tau)$, which encodes the Bernoulli-rational coboundary correction.*

j	ε_j	c_S^j	$k_S^j(\tau)$	$k_T^j(\tau)$
1	−	4	-10τ	$-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$
2	+	1	$45\tau^2$	$-9\tau + \frac{45}{2}\tau^2 - 5\tau^3 - 21\tau^5 + 30\tau^7 - \frac{45}{2}\tau^8 + 5\tau^9$
3	−	−25	$-120\tau^3$	$40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$
4	+	−3	$210\tau^4$	$12\tau - 105\tau^3 + 105\tau^4 + 63\tau^5 - 105\tau^6 + 30\tau^7$
5	−	42	$-252\tau^5$	$-42\tau^2 + 210\tau^4 - 252\tau^5 + 84\tau^6$
6	+	3	$210\tau^6$	$-12\tau + 105\tau^3 - 105\tau^4 - 63\tau^5 + 105\tau^6 - 30\tau^7$
7	−	−25	$-120\tau^7$	$40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$
8	+	−1	$45\tau^8$	$9\tau - \frac{45}{2}\tau^2 + 5\tau^3 + 21\tau^5 - 30\tau^7 + \frac{45}{2}\tau^8 - 5\tau^9$
9	−	4	$-10\tau^9$	$-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$

Table 1: The nine universal polynomial-kernel pairs (k_S^j, k_T^j) of Theorem 4.1, with Brown’s basis values $c_S^j = [P_S^{\varepsilon_j}]_{X^{10-j}Y^j}$. Computed exactly via δ_T^{-1} in sympy rationals [12]. Pairs (1, 9) and (3, 7) share k_T identically; pairs (2, 8) and (4, 6) share k_T up to overall sign (matching the sign flip in c_S^j); $j = 5$ is self-dual.

5 Specialisation to Δ and $\widehat{\Delta}$

We now specialise Theorem 1.1 to $f = \Delta$ and $f = \widehat{\Delta}$ and verify Corollaries 1.3, 1.4 numerically.

Corollary 5.1 (Explicit formula for $\omega^\pm(\Delta)$). *For $\Delta \in S_{12}$, any $\tau_0 \in \mathbb{H}$, and any interior $j \in \{1, \dots, 9\}$,*

$$\omega^{\varepsilon_j}(\Delta) = \frac{(2\pi i)^{11}}{c_S^j} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) k_T^j(\tau) d\tau \right]. \quad (16)$$

Corollary 5.2 (Explicit formula for $\eta^\pm(\widehat{\Delta})$). *For $\widehat{\Delta} \in S_{12}^!$, any $\tau_0 \in \mathbb{H}$, and any interior $j \in \{1, \dots, 9\}$,*

$$\eta^{\varepsilon_j}(\widehat{\Delta}) = \frac{(2\pi i)^{11}}{c_S^j} \left[\int_{-1/\tau_0}^{\tau_0} \widehat{\Delta}(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \widehat{\Delta}(\tau) k_T^j(\tau) d\tau \right]. \quad (17)$$

The kernels in (16) and (17) are *identical*; only the modular form f under the integral changes.

Numerical verification. At $\tau_0 = 0.3 + 1.2i$, the integrals in (16) and (17) are computed at 45-digit `mpmath` precision in [9]. Defining the DR-regularised L -value $L_{\text{reg}}(f, j+1)$ by $\omega^{\varepsilon_j}(f)c_S^j/[(2\pi)^{10-j}(-i)^j \binom{10}{j} j!]$ (which agrees with the classical $L(\Delta, j+1)$ for $f = \Delta$, by Lemma 5.3 below):

j	ε_j	$L_{\text{reg}}(\Delta, j+1)$	extracted $\omega^{\varepsilon_j}(\Delta)$
1	−	0.14637454209126598...	−5 585 015.379310401 ... i
2	+	0.31524156588099308...	−68 916 772.80959519 ...
3	−	0.50161764751090221...	−5 585 015.379310401 ... i
4	+	0.66670918843400364...	−68 916 772.80959519 ...
5	−	0.79212283864603056...	−5 585 015.379310401 ... i
6	+	0.87735412538866091...	−68 916 772.80959519 ...
7	−	0.93070703029812609...	−5 585 015.379310401 ... i
8	+	0.96212645969442586...	−68 916 772.80959519 ...
9	−	0.97980908825122051...	−5 585 015.379310401 ... i

All nine rows lockstep to relative residual $< 6 \times 10^{-42}$ [11]. The same nine integrals with $\hat{\Delta}$ in place of Δ produce η^+, η^- in lockstep to relative residual $\sim 10^{-17}$ [13], agreeing with Brown's reported values to all 20 reported digits (Corollary 1.4); the slightly reduced precision reflects the larger absolute magnitudes ($|\eta^\pm| \sim 10^{11}$ versus $|\omega^\pm| \sim 10^7$).

Lemma 5.3 (Mellin-transform recovery for Δ). *As $\tau_0 \rightarrow i\infty$, the T -segment integral in (16) vanishes (since all $M_j(\Delta, \tau_0) \rightarrow 0$ by cuspidality of Δ), the S -segment becomes $[0, i\infty]$, and the Mellin transform $\int_0^{i\infty} \Delta(\tau) \tau^j d\tau = i^{j+1} j! / (2\pi)^{j+1} \cdot L(\Delta, j+1)$ identifies $L_{\text{reg}}(\Delta, j+1) = L(\Delta, j+1)$ with the standard critical L -value. (16) then collapses to the classical Eichler–Shimura formula $\omega^{\varepsilon_j}(\Delta) = (2\pi)^{10-j} (-i)^j \binom{10}{j} j! L(\Delta, j+1) / c_S^j$.*

Proof. $M_j(\Delta, \tau_0) \rightarrow 0$ exponentially because the T -path integrand $\Delta(\tau) \tau^j$ decays as $|q|^1 = e^{-2\pi \text{Im}(\tau)}$ on the segment. Hence $P \rightarrow 0$ by Proposition 3.2, the T -segment contribution to (16) drops to zero, and the S -segment $[-1/\tau_0, \tau_0]$ stretches to $[0, i\infty]$. The Mellin identification then follows from the Hecke–Weil integral representation of $L(\Delta, s)$. $\square$

For $\hat{\Delta}$ the $\tau_0 \rightarrow i\infty$ limit is unavailable — both integrals individually diverge as the q^{-1} Fourier mode amplifies each $|q_0|^{-1} = e^{2\pi \text{Im}(\tau_0)} \rightarrow \infty$. But for finite $\tau_0 \in \mathbb{H}$ both integrals are finite and (17) yields $\eta^\pm(\hat{\Delta})$. The basepoint is the regulator.

Remark 5.4 (Exponentially convergent q -series expansion). *Substituting the Fourier expansion $f(\tau) = \sum_{n \geq -N_f} a_f(n) e^{2\pi i n \tau}$ into (16) (resp. (17)) and integrating term-by-term via the elementary identity $\int e^{2\pi i n \tau} \tau^j d\tau = \sum_{k=0}^j \frac{(-1)^k j!}{(j-k)!(2\pi i n)^{k+1}} e^{2\pi i n \tau} \tau^{j-k}$ gives an exponentially convergent series for the periods:*

$$\omega^{\varepsilon_j}(f) = \frac{(2\pi i)^{11}}{c_S^j} \sum_{n \neq 0} a_f(n) \mathcal{K}^j(n, \tau_0) + (\text{constant-term piece, vanishing for } f \in S_k^!), \quad (18)$$

where $\mathcal{K}^j(n, \tau_0)$ is an explicit elementary function of n and τ_0 built from finite polynomial combinations of $q_0^n = e^{2\pi i n \tau_0}$, $\tilde{q}_0^n = e^{-2\pi i n / \tau_0}$, τ_0^a , $(\tau_0 - 1)^a$, $(-1/\tau_0)^a$, and inverse powers $1/(2\pi i n)^{b+1}$. Since $|q_0|, |\tilde{q}_0| < 1$ for every $\tau_0 \in \mathbb{H}$, the series converges super-exponentially in $|n|$.

For $f = \Delta$, only $n \geq 1$ contributes with $a_f(n) = \tau(n)$, yielding an exponentially convergent series for the classical $\omega^\pm(\Delta)$ — the explicit linear functional of $\{\tau(n)\}_{n \geq 1}$ promised in the introduction. For $f = \hat{\Delta}$, the $n = -1$ term contributes one extra finite piece $q_0^{-1} \mathcal{K}^j(-1, \tau_0)$; the $n \geq 2$ terms decay super-exponentially in n despite the $a_{\hat{\Delta}}(n) \sim e^{4\pi \sqrt{n}}$ coefficient growth. Comparing the Δ -version of (18) to BFK's incomplete-gamma expansion of $L(\Delta, s)$ [3] provides a concrete sanity check; writing the same expansion for $\eta^\pm(\hat{\Delta})$ produces the natural DR-analog of BFK's regularised L -value series, valid uniformly across $S_{12}^!$. Detailed term-by-term comparison of these series is deferred to a follow-up note.

6 Cohomology and basepoint independence

Theorem 6.1 (Cohomological well-definition). *The class $[C^f] \in H^1(\Gamma; V_{10} \otimes \mathbb{C})$ is independent of τ_0 . For $f \in S_{12}^!$, this class is cuspidal, and Brown's decomposition (12) reads off cohomologically-meaningful coordinates: $\omega^+(f) = \text{coordinate of } [C^f] \text{ in } [P^+]$, $\omega^-(f) = \text{coordinate in } [P^-]$.*

Proof. For $\tau_0, \tau_1 \in \mathbb{H}$, define the basepoint-shift potential

$$h(X, Y; \tau_0, \tau_1) := (2\pi i)^{11} \int_{\tau_0}^{\tau_1} f(\tau) (X - \tau Y)^{10} d\tau \in V_{10} \otimes \mathbb{C}. \quad (19)$$

The same path-additivity / modularity calculation as in Lemma 2.2, applied to the basepoint-shifted integrals, gives

$$C_\gamma^f(\tau_1) - C_\gamma^f(\tau_0) = -[h(\tau_0, \tau_1)|_\gamma - h(\tau_0, \tau_1)], \quad (20)$$

manifestly a coboundary in $V_{10} \otimes \mathbb{C}$. Hence $[C_\bullet^f]$ is independent of τ_0 . Cuspidality of the class for $f \in S_{12}^!$ follows from Proposition 3.1. The basis-coordinate identification is (12). $\square$

Corollary 6.2 (τ_0 - and j -independence in Theorem 1.1). *The right side of (2) is independent of $\tau_0 \in \mathbb{H}$ (Theorem 6.1) and of the choice of interior $j \in \{1, \dots, 9\}$ within its parity sector (since all interior- j extractions read off the same basis coordinate of $[C^f]$ in (12)). Together with Theorem 4.1, this completes the proof of Theorem 1.1. Numerically, $|\Delta\omega^\pm| < 3 \times 10^{-38}$ and $|\Delta\eta^\pm| < 1.3 \times 10^{-32}$ between $\tau_0 = 0.3 + 1.2i$ and $\tau_0 = -0.2 + 0.9i$ [9].*

Remark 6.3 ($C_S'^f$ in Zagier's basis). *Using (12) together with $P_S^+ = -\frac{36}{691}P_0 + P_1$, $P_S^- = P_2$:*

$$C_S'^f = a_0(f)P_0 + \omega^+(f)P_1 + \omega^-(f)P_2,$$

with $a_0(f) = c(f) - \frac{36}{691}\omega^+(f) = [C_S'^f]_{X^{10}}$ the convention-dependent Eisenstein coordinate. At $\tau_0 = 0.3 + 1.2i$ [13]:

$$\begin{aligned} C_S'^\Delta &= (2.752 \times 10^5 - 6.156 \times 10^5 i)P_0 + (-68\,916\,772.80959519 \dots)P_1 + (-5\,585\,015.379310402 \dots i)P_2, \\ C_S'^{\hat{\Delta}} &= (-1.532 \times 10^{11} + 6.163 \times 10^{11} i)P_0 + (127\,202\,100\,647.1771 \dots)P_1 + (10\,276\,732\,343.64913 \dots i)P_2. \end{aligned}$$

The P_1, P_2 coordinates are precisely $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$.

7 Comparison with the BFK regularisation

Bringmann–Fricke–Kent [3] regularise the divergent integral $\int_0^{i\infty} f(\tau) \tau^{s-1} d\tau$ for weakly holomorphic $f \in S_k^!$ via a non-holomorphic exponential factor $e^{2\pi\nu \text{Im}(\tau)}$ continued in ν . At integer $s \in \{1, \dots, k-1\}$ their regularised L^* -value reduces to the absolutely-convergent incomplete-gamma sum

$$L_{\text{BKF}}^*(f, s) = \sum_{n \neq 0} a_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (21)$$

where the sum extends over *all* nonzero Fourier modes, including the negative-index principal-part modes of weakly holomorphic forms. At integer s the incomplete-gamma function reduces to a polynomial in its second argument ($\Gamma(s, x) = (s-1)! e^{-x} \sum_{r < s} x^r / r!$), so each $n < 0$ contribution $a_f(n) \Gamma(s, -2\pi|n|) / (-2\pi|n|)^s$ is a finite complex number and (21) is a well-defined finite sum. Zagier's formula then assembles these eleven L^* -values into a polynomial $r_{\text{BKF}}(f) \in V_{k-2} \otimes \mathbb{C}$.

Proposition 7.1 (r_{BFK} and the DR-B representative agree). *Let $f \in \{\Delta, \hat{\Delta}\}$. Then:*

1. $r_{\text{BFK}}(f) \in W_{12}$: both period relations $r_{\text{BFK}}(f)|(1+S) = 0$ and $r_{\text{BFK}}(f)|(1+U+U^2) = 0$ (with $U = TS$, weight -10 slash action) hold to relative residual $\sim 10^{-48}$ at 50-digit *mpmath* precision [14].
2. The polynomial- V_{10} DR-B representative $C_S^{f'}(\tau_0)/(2\pi i)^{11}$ of Theorem 1.1 agrees with $r_{\text{BFK}}(f)$ exactly on the nine interior monomials $X^1, \dots, X^9$ (relative residual $< 10^{-44}$). At the boundary monomials X^0, X^{10} the two polynomials differ by an Eisenstein-direction term $c(\tau_0)(1 - X^{10})$; the constant c is τ_0 -dependent (reflecting the boundary freedom in the polynomial- V_{10} DR-B representative noted after $(\star)$) and at $\tau_0 = 0.3 + 1.2i$ has magnitude $\approx 5.6 \times 10^{-3}$ for $f = \Delta$ and $\approx 1.0 \times 10^3$ for $f = \hat{\Delta}$.
3. The Hurwitz-zeta-extended DR-B representative $r_{\text{DR}}^{\tilde{k}_T}(f)$ of §8 agrees with $r_{\text{BFK}}(f)$ at all eleven monomials $X^0, \dots, X^{10}$ to relative residual $\sim 10^{-45}$, for both $f = \Delta$ and $f = \hat{\Delta}$. Equivalently, $\Lambda_{\text{BFK}}^*(f, s) = \Lambda_{\text{DR}}^*(f, s)$ at all integer $s \in \{1, \dots, 11\}$, and (numerically) at the non-integer test points $s \in \{-0.3, 0.5, 1.5, \dots, 6.5\}$ and complex test points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$ for both f , at relative residual $\sim 10^{-47}$ [14]. The Hurwitz-zeta T -kernel is the DR-B representative that pins the boundary coefficients down τ_0 -canonically.

What distinguishes the two constructions, then, is not the value of the period polynomial they deliver (they deliver the same one) but their geometry. BFK uses two infinite contours $\int_0^{i\infty}$ and $(\int_0^{i\infty})|S$ with a non-holomorphic regulator inserted at both endpoints; DR-B uses two *finite* contours $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$ that stay strictly inside $\mathbb{H}$ and require no regulator. The finite-contour construction is what enables the natural extension to meromorphic forms with interior poles (§9): a regulator-based scheme has no immediate analog there because the cusp-exchange symmetry that pins down BFK's regulator at the cusps has no analog at an interior pole.

8 The complex- s extension: a finite-contour L -function via Hurwitz zeta

Theorem 1.1 extracts the period-polynomial coefficients of $f \in S_{12}^1$ at the nine interior integer monomials $j \in \{1, \dots, 9\}$ via finite-contour integrals on the DR-B contour pair $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$. For holomorphic $f = \Delta$ we now extend the same construction to general complex s , producing a closed-form expression for the completed L -function $\Lambda^*(\Delta, s) := (2\pi)^{-s}\Gamma(s)L(\Delta, s)$ as an integral pair over the *same* DR-B contour, with the polynomial T -kernel k_T^j of Table 1 replaced by a Hurwitz-zeta function of τ .

The gap identity. For $\Delta \in S_{12}$ and any $\tau_0 \in \mathbb{H}$, splitting the canonical Mellin integral $\int_0^{i\infty} \Delta(\tau) \tau^{s-1} d\tau$ at the DR-B endpoints $\{-1/\tau_0, \tau_0\}$ and using S -modularity of Δ on the $[0, -1/\tau_0]$ piece (via $\tau \mapsto -1/\tau$) gives, for all $s \in \mathbb{C}$,

$$\Lambda^*(\Delta, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} \tau(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(12-s, -2\pi i n \tau_0)}{(2\pi n)^{12-s}} \right]. \quad (22)$$

The second piece is the BFK incomplete-gamma sum of [3] *split at τ_0 rather than at $\tau = i$* : at $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (22) reduces to the standard BFK formula. Both pieces on the right of (22) are individually entire in s and depend on τ_0 , but their sum equals the τ_0 -invariant $\Lambda^*(\Delta, s)$. Identity (22) is verified numerically to relative residual $\sim 10^{-54}$ at $\tau_0 = 0.3 + 1.2i$, 45-digit precision, for $s = 1, \dots, 11$ [14].

The τ_0 -invariance condition. We seek a kernel $\tilde{k}_T(\tau, s)$ such that the BFK-split-at- τ_0 sum in (22) is replaced by a finite-contour integral $\int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau$ over the same T -segment used in Theorem 1.1. Differentiating the candidate

$$\Phi_s(\tau_0) := i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau \right]$$

in τ_0 , using $\Delta(-1/\tau_0) = \tau_0^{12} \Delta(\tau_0)$ for the S -segment endpoint and $\Delta(\tau_0 - 1) = \Delta(\tau_0)$ for the T -segment endpoint, the requirement $d\Phi_s/d\tau_0 \equiv 0$ reduces (canceling the common factor $i^{-s} \Delta(\tau_0)$) to

$$\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{11-s} \quad \text{for all } \tau \in \mathbb{H}. \quad (23)$$

This is the complex- s analog of the polynomial coboundary equation ($\star$): it asks for an inverse of δ_T on the two-term right-hand side.

Hurwitz zeta solves (23). The Hurwitz raising identity $\zeta(\sigma, a) - \zeta(\sigma, a+1) = a^{-\sigma}$, applied at $(\sigma, a) = (1-s, \tau)$ and $(s-11, \tau)$, gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-11, \tau+1) - \zeta(s-11, \tau) = -\tau^{11-s}.$$

Combining with the sign $e^{i\pi(s-1)}$ from (23):

Theorem 8.1 (Complex- s finite-contour L -function for Δ). *For all $s \in \mathbb{C} \setminus \{1, 12\}$ and all $\tau_0 \in \mathbb{H}$,*

$$\Lambda^*(\Delta, s) = i^{-s} \left[\int_{-1/\tau_0}^{\tau_0} \Delta(\tau) \tau^{s-1} d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (24)$$

where the Hurwitz-zeta T -kernel is

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-11, \tau+1). \quad (25)$$

Both contour integrals are finite for every $\tau_0 \in \mathbb{H}$, with no regularisation, analytic continuation, or limit required. At integer $s = j+1 \in \{1, 2, \dots, 11\}$, both ζ -terms in (25) collapse to Bernoulli polynomials in τ via the standard identity $\zeta(-n, a) = -B_{n+1}(a)/(n+1)$, and $\tilde{k}_T(\tau, j+1)$ becomes a polynomial in τ . For the nine interior values $j \in \{1, \dots, 9\}$ this polynomial agrees (up to the trivial rescaling $(-1)^j \binom{10}{j}$ that converts the $\omega^\pm$ -normalised kernel to the L -value-normalised one) with the polynomial T -kernel k_T^j of Table 1. The two boundary cases $s = 1$ and $s = 11$ — which the bounded-degree V_{10} Bernoulli inversion of §3 cannot fix (the convention $p_{10} = 0$ leaves an unresolved Eisenstein direction there) — are recovered seamlessly in (24) because the Hurwitz inversion does not need a polynomial-degree truncation.

Proof. Both sides of (24) are meromorphic in s , with poles on the right side only from the explicit Hurwitz factors at $s = 1, 12$ (poles cancelled, in the sum with the S -segment, by the analytic structure of Λ^*). By construction (23), the right side of (24) is τ_0 -invariant: differentiating in τ_0 reduces to identity (23), which holds for $\tilde{k}_T$ from (25) by the Hurwitz raising identity applied to each ζ term separately. Hence the right side equals its $\tau_0 \rightarrow i\infty$ limit: the T -segment integral $\int_{\tau_0-1}^{\tau_0} \Delta \tilde{k}_T d\tau$ vanishes exponentially because Δ does and $\tilde{k}_T$ grows at most algebraically in τ , while the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0} \Delta \tau^{s-1} d\tau$ approaches $i^{-s} \int_0^{i\infty} \Delta \tau^{s-1} d\tau = \Lambda^*(\Delta, s)$ by the Hecke–Weil Mellin identity (extending Lemma 5.3 from integer j to complex s). This identifies the right side with $\Lambda^*(\Delta, s)$. $\square$

Numerical verification. Formula (24) reproduces the standard BFK-at- $\tau = i$ incomplete-gamma value of $\Lambda^*(\Delta, s)$ to relative residual $\sim 10^{-53}$ at 45-digit `mpmath` precision [14], at every integer $s \in \{1, 2, \dots, 11\}$ (including the two boundaries $s = 1, 11$ unreachable by Theorem 1.1), every half-integer test point $s \in \{-0.3, 0.5, 1.5, 2.5, \dots, 6.5\}$, and every complex test point $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$. Basepoint-independence is verified directly: at fixed $s = 3 + 1.5i$, evaluating the right side of (24) at $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4 + 0.8i, 0.2 + 2i\}$ gives one value to relative residual $\sim 10^{-53}$.

Structural interpretation. The polynomial T -kernels k_T^j of Theorem 4.1 are the integer- s specialisations of $\tilde{k}_T(\tau, s)$. The bounded-degree Bernoulli inversion on V_{10} in §3 is the integer-weight shadow of the analytic Hurwitz raising identity

$$\delta_T[\zeta(1-s, \tau+1)] = \zeta(1-s, \tau+2) - \zeta(1-s, \tau+1) = -(\tau+1)^{s-1},$$

which inverts δ_T uniformly at all complex s without bounding the polynomial degree. In cohomological language, the finite-dimensional coefficient system V_n of Eichler–Shimura extends to an analytic family of principal-series-type coefficient systems on $\mathbb{H}$ parametrised by s , and Theorem 8.1 is the explicit DR-B representative for Δ in this larger system. This is the holomorphic, finite-contour analog of the Lewis–Zagier period-function framework for Maass wave forms [6].

Same formula for Δ and $\hat{\Delta}$ at all s . The right side of (24) is finite for any $f \in S_{12}^!$ in place of Δ , and indeed for any meromorphic modular form of weight 12 with no pole on the DR-B contour pair — both segments stay strictly inside $\mathbb{H}$, and the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ is analytic in $\tau \in \mathbb{H}$. Applying (24) to $\hat{\Delta}$ produces $\Lambda_{\text{DR}}^*(\hat{\Delta}, s)$ for every $s \in \mathbb{C}$; at integer $s \in \{1, \dots, 11\}$ this equals the BFK L^* -value of (21) (Proposition 7.1), so that $L^{\text{DR}}(\hat{\Delta}, s) = L_{\text{BFK}}^*(\hat{\Delta}, s)$ at all eleven integer points including the boundaries $s = 1, 11$. Equation (24) thereby gives the canonical extension of $L_{\text{BFK}}^*(\hat{\Delta}, \cdot)$ to all $s \in \mathbb{C}$, as a finite-contour integral pair with no regulator and no analytic continuation step. Working out the non-integer behaviour of $L^{\text{DR}}(\hat{\Delta}, s)$ explicitly, and the analog of the functional equation $\Lambda^*(s) = i^k \Lambda^*(k-s)$ in this finite-contour form, is the next analytic question.

9 Outlook: meromorphic forms, and a gauge-invariance analogy

Recap. We have given (i) an explicit linear functional $f \mapsto (\omega^+(f), \omega^-(f))$ on $S_{12}^!$ via the finite-endpoint DR-B cocycle (Theorem 1.1); (ii) its extension to a finite-contour formula for the completed L -function $\Lambda^*(f, s)$ at every complex s via the Hurwitz-zeta T -kernel (Theorem 8.1); and (iii) the equality of this L -function with the BFK regularised L -function at all integer s for $f \in \{\Delta, \hat{\Delta}\}$ (Proposition 7.1). The DR-B construction is therefore not a competitor to BFK; it is a manifestly finite reformulation of the same object, replacing BFK’s non-holomorphic regulator at the two cusps with two finite line segments inside $\mathbb{H}$.

Meromorphic forms with interior poles. The advantage of the finite-contour formulation appears once f acquires interior poles, where BFK has no analog: the cusp-exchange symmetry $\tau \rightarrow -1/\tau$ that pins BFK’s regulator at the two cusps has no counterpart at an interior $\tau_p \in \mathbb{H}$. The DR-B construction, by contrast, requires only that the contour pair $\{[-1/\tau_0, \tau_0], [\tau_0 - 1, \tau_0]\}$ avoid the poles of f . Choosing any τ_0 off the Γ -orbit of the polar locus $f^{-1}(\infty) \cap \mathbb{H}$, the formula (24) delivers $\Lambda_{\text{DR}}^*(f, s)$ for arbitrary meromorphic f of weight 12 at every $s \in \mathbb{C}$.

What changes is the τ_0 -dependence. For $f \in S_{12}^!$, $\Lambda_{\text{DR}}^*(f, s)$ is τ_0 -invariant by Theorem 8.1. For meromorphic f , the basepoint-shift identity (20) acquires residue contributions whenever the contour crosses a pole. Explicitly, for a simple pole of f at $\tau_p \in \mathbb{H}$, basepoint-shift across τ_p adds

$$2\pi i (2\pi i)^{11} \text{Res}(f, \tau_p) (X - \tau_p Y)^{10} \in V_{10} \otimes \mathbb{C} \quad (26)$$

to the cocycle polynomial C^f , and the analogous expression with derivatives in τ_p of $(X - \tau_p Y)^{10-i}$, $i = 0, \dots, m-1$, for a pole of order m . The presentation of $\Lambda_{\text{DR}}^*(f, s)$ as a finite-contour integral therefore becomes τ_0 -piecewise constant on the connected components of $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with discrete *wall-crossing* jumps (26) at the Γ -orbits of the poles. This wall-crossing structure is a feature of the meromorphic L -function, not a defect of any one representative: it encodes the residue data of f in a form that is not visible in BFK’s regulator-based construction (because that construction cannot see interior poles in the first place).

Working out the explicit residue polynomials for canonical meromorphic forms (e.g. $1/E_6$ with simple poles at $\tau = i$ and its Γ -orbit; $1/E_4$ at $\tau = \rho$), and the arithmetic of the resulting L -functions and their wall-crossing data, is the natural successor program.

A gauge-invariance analogy, and an open question. The cocycle C^f carries more data than the cohomologically-meaningful periods $\omega^\pm(f)$: an entire coboundary direction δP must be subtracted, and the choice of representative P (parametrised by the unconstrained $p_{10} \in \mathbb{C}$) is irrelevant for the final answer. This carries strong echoes of off-shell gauge data in Feynman amplitudes with massless gauge bosons — longitudinal polarisations, Faddeev–Popov ghosts — which appear in the intermediate bookkeeping but cancel in physical (gauge-invariant) observables. In QFT, the on-shell S-matrix programme circumvents the gauge problem by building amplitudes directly from gauge-invariant on-shell building blocks (helicity variables, BCFW recursion, unitarity cuts [7, 8]), so the gauge-redundant intermediate data never enters: one is forced into “fix the gauge” only because one has made the upstream choice to work off-shell.

The analog question for periods: does there exist a “manifestly cohomological” construction that produces $\omega^\pm(f)$ directly from the Fourier coefficients of f , sidestepping the intermediate cocycle C^f and coboundary δP scaffolding entirely? The exponentially convergent q -coefficient series of Remark 5.4 is a partial step — input $\{a_f(n)\}$, output $\omega^\pm$, no C^f or P visible in the final formula — but the kernels appearing in that series were themselves derived via δ_T^{-1} , so the cocycle structure remains implicit. A formulation in which the kernels are constructed without going through the cocycle / coboundary intermediate stage — i.e. the modular-form analog of writing down on-shell amplitudes directly via BCFW or unitarity rather than gauge-fixing an off-shell Lagrangian — would seem to require an independent characterisation of the cuspidal-cohomology coordinates of $H_{\text{cusp}}^1(\Gamma; V_n)$. We flag this as an open question; what the modular-form analog of the on-shell S-matrix programme might be is, in our view, a worthy direction.

References

- [1] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [2] N. Diamantis and L. Rolin, *Period polynomials, derivatives of L-functions, and zeros of polynomials*, arXiv:1707.04814 (2017).
- [3] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special L-values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.

- [4] D. A. McGady, *L-functions for meromorphic modular forms*, arxiv:1806.abcd
- [5] W. Kohnen and D. Zagier, *Modular forms with rational periods*, in *Modular Forms* (R.A. Rankin, ed.), Ellis Horwood, 1984, pp. 197–249.
- [6] J. Lewis and D. Zagier, *Period functions for Maass wave forms. I*, Annals of Mathematics, 153 (2001), 191-258.
- [7] Y.-t. Huang and D. McGady, *Consistency Conditions for Gauge Theory S Matrices from Requirements of Generalized Unitarity*, Phys. Rev. Lett. **112** (2014) 241601, arXiv:1307.4065 [hep-th].
- [8] W.-M. Chen, Y.-t. Huang, and D. A. McGady, *Anomalies without an action*, arXiv:1402.7062 [hep-th] (2014).
- [9] `period_extraction_DR_general.ipynb` (executed Jupyter notebook). Companion source file; implements `extract_periods(coeffs_dict, tau0, k)`: the single q -series-agnostic function reproducing the $\omega^\pm, \eta^\pm$ values of Corollaries 1.3, 1.4 and the basepoint-independence check of Corollary 6.2, at 45-digit `mpmath` precision.
- [10] `verify_bernoulli_formula.py`, companion script. Confirms agreement of the closed-form Bernoulli operator (10) with the upper-triangular recursion of Proposition 3.1 on every p_m , $m = 0, \dots, 9$, to relative residual $\sim 10^{-42}$ at 45-digit working precision.
- [11] `verify_omega_plus_formula.py`, companion script. Evaluates each of the nine equivalent forms of $\omega^\pm(\Delta)$ in (16) by computing $L(\Delta, s)$ via the BFK incomplete-gamma sum, confirming agreement with Brown’s reported values to relative error $< 6 \times 10^{-42}$. Source of the $L_{\text{reg}}(\Delta, s)$ table in §5.
- [12] `compute_period_kernels.py`, companion script. Computes the nine universal $\mathbb{Q}$ -polynomial kernel pairs (k_S^j, k_T^j) of Table 1 exactly via sympy rationals, by applying the δ_T^{-1} Bernoulli operator monomial-by-monomial as in (15).
- [13] `compute_numerical_polynomials.py`, companion script. Computes the explicit modified co-cycle values C_S^{fj} of Remark 6.3 at $\tau_0 = 0.3 + 1.2i$ for $f \in \{\Delta, \hat{\Delta}\}$ at 30-digit `mpmath` precision. Verifies period extraction from all nine interior monomials to lockstep agreement.
- [14] `dr_b_Lfunction_complex_s.py`, companion script. Computes $\Lambda^*(\Delta, s)$ both via the BFK incomplete-gamma reference sum and via the Hurwitz-zeta DR-B finite-contour functional of Theorem 8.1. Verifies (i) the gap identity (22) for $s = 1, \dots, 11$ at relative residual $\sim 10^{-54}$; (ii) the integer- s reduction of Theorem 8.1 at all $s \in \{1, \dots, 11\}$ including the two boundary values $s = 1, 11$ unreachable by the bounded-degree V_{10} Bernoulli inversion; (iii) the non-integer test points $s \in \{-0.3, 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, 6, 6.5\}$ and the complex test points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$; and (iv) basepoint independence at fixed $s = 3 + 1.5i$ across $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4 + 0.8i, 0.2 + 2i\}$. All checks agree at relative residual $\sim 10^{-53}$ at 50-digit `mpmath` precision.
- [15] Long-form companion document `analytic_periods_weight12.tex / .pdf`. Same mathematical content as the present note with extensive pedagogical discussion of cocycles, coboundaries, the Bernoulli-operator derivation, and worked numerical polynomials. The present note is the math-paper-style restatement with (2) as the front matter.